

| Question | Marking details                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             | Marks available |     |     |       |       |      |
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|          |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             | AO1             | AO2 | AO3 | Total | Maths | Prac |
| 7        | <ul style="list-style-type: none"> <li>• <b>Extract</b> mRNA from the bacterium (which is synthesising the proteins).</li> <li>• Use <b>reverse transcriptase</b> to synthesise a {single strand of DNA/ cDNA} using the mRNA (as a template).</li> <li>• Use <b>DNA polymerase</b> to produce double stranded DNA.</li> <li>• <b>Sticky ends</b> are added to the end of each DNA strand of the DNA.</li> <li>• The DNA is copied many times using the <b>PCR</b>.</li> <li>• Plasmids from bacteria are isolated</li> <li>• <b>Restriction {endonuclease/ enzymes}</b> is used to open the plasmid (at a specific point)</li> <li>• producing <b>sticky ends complementary</b> to the DNA.</li> <li>• <b>DNA ligase</b> is used to bind the DNA into the plasmid.</li> <li>• <b>Antibiotic resistance genes / markers/ LacZ</b> are used to select bacteria containing the recombinant plasmids.</li> </ul> <p>Concerns:</p> <ul style="list-style-type: none"> <li>• dispersal of pollen from engineered crops to wild relatives/ contamination of organic crops</li> <li>• unknown effects of eating new protein which is toxic to insects/ health concerns from eating GM crop</li> <li>• a reduction in biodiversity e.g. beneficial insects killed/ affect on food chains</li> <li>• Could result in resistance to protein in insects/ selection pressure</li> </ul> |                 |     |     |       |       |      |